

"I'm Basically a 24-Hour Navigator": A Case Study of PrEP Navigators' Use of LINE Messaging Application for Case Management in Taiwan

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Abstract

This case study examines how PrEP navigators in Taiwan adopt the LINE messaging application to support preventive care work beyond clinical settings. Based on in-depth interviews with six PrEP navigators, we analyze how LINE is embedded in everyday case management practices, including appointment follow-up and responding to emergent concerns outside the clinic. We identify four key affordances of LINE that enable navigators to flexibly adapt care to clients' situations: universality, multimodality, asynchrony, and private one-on-one communication. By foregrounding navigators' situated practices, this paper contributes an empirical account of mainstream messaging platforms in frontline preventive care, discusses tensions between technological convenience and care burden, and proposes directions for the design of future digital health systems.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

PrEP care, PrEP navigator, Instant messaging, Digital health

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1 Introduction

Pre-exposure prophylaxis (PrEP) is a preventive HIV medication that is taken by individuals who are not currently ill but remain at ongoing risk of infection [2, 11, 29]. As a form of preventive care, PrEP requires sustained engagement and medication adherence [7, 20]. In many settings, these responsibilities are carried out by PrEP case managers or navigators, who support clients' engagement

through ongoing communication between healthcare providers and clients [25, 35]. According to Taiwan CDC, PrEP navigators support clients from initial linkage and suitability assessment through ongoing follow-up, including adherence check-ins, periodic testing, and prevention counseling and education. They may also coordinate referrals when additional needs arise, such as substance-use or mental health support [27].

Prior research in HCI and digital health has examined the use of technologies to support PrEP adherence and healthcare [4, 13, 16, 17, 26, 33]. More recently, AI-powered conversational systems have been discussed as potential tools for supporting healthcare delivery including patient engagement and health education [1, 12, 14, 23, 34]. Technologies introduced without careful attention to established practices often face significant adoption challenges in real-world settings [6, 21, 24]. Understanding how healthcare workers currently leverage available communication tools is a necessary step for informing the design of future health technologies.

In Taiwan, LINE is the most widely used instant messaging platform, with approximately 93% of the population actively using it across age groups [19]. In the context of PrEP case management, some navigators adopt LINE as a primary channel for communicating with clients, despite the platform not being designed for clinical or case management purposes. This makes Taiwan a valuable case for examining how a mainstream messaging platform is adopted to support preventive healthcare work in practice.

The aim of this study is to understand how PrEP navigators in Taiwan use LINE to support case management in their everyday practice. Accordingly, we address two research questions: **(1) What do PrEP navigators in Taiwan use LINE to do in their case management work?** **(2) What affordances of LINE support these practices?** By examining this context-specific case, we provide an empirical account of messaging platform use in frontline preventive care and offer insights relevant to the design of future digital support tools.

2 Method

2.1 Participants

We recruited six healthcare professionals currently working as PrEP navigators in Taiwan (Table 1). All participants had direct experience supporting PrEP clients as part of their clinical or public health roles. To capture variation in practice contexts, we aimed to include participants from different geographic regions (northern, central, and southern Taiwan). Participants were recruited via email invitations distributed through a centralized contact list maintained

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Table 1: Participant Information. Participants (2 males, 4 females) were drawn from northern ($n = 3$), central ($n = 2$), and southern ($n = 1$) Taiwan and held a range of roles, including case managers, HIV case managers, project assistants, and research assistants. All participants had multiple years of experience working as PrEP navigators (3–4 years). The scale of PrEP care varied substantially across participants, with reported caseloads ranging from several dozen clients to more than 1,800 clients.

No.	Gender	Hospital Location	Job Title	Years as Navigator	Number of PrEP Clients
P1	Male	Northern Taiwan	Project Assistant	3	More than 1,800
P2	Male	Northern Taiwan	Case Manager	4	Hundreds (exact number unknown)
P3	Female	Northern Taiwan	Case Manager	4	20
P4	Female	Central Taiwan*	HIV Case Manager	4	Co-manage 172 clients with a team
P5	Female	Central Taiwan*	HIV Case Manager	3	Approximately 100
P6	Female	Southern Taiwan	Research Assistant	3	40

*P4 and P5 were affiliated with the same institution.

by a national PrEP program in Taiwan. This contact list included email addresses of PrEP navigators affiliated with PrEP programs across different regions of Taiwan, allowing us to reach eligible participants directly. Recruitment criteria prioritized navigators with substantial hands-on experience in PrEP-related care and case management. Participation was voluntary and participants received compensation of approximately USD 63.30 (TWD 2,000).

2.2 Data Collection and Analysis

We conducted one-on-one, semi-structured, in-depth interviews with six participants. Interviews lasted approximately 90–120 minutes and were conducted either in person (P2) or online (P1, P3–P6) based on participants' preferences and availability. All interviews were conducted in Mandarin Chinese, participants' native language, audio-recorded, transcribed verbatim, and subsequently translated into English for analysis and reporting in this paper.

We analyzed the transcripts using thematic analysis approach [5, 28]. First, three researchers independently read and re-read the transcripts to familiarize themselves with the data. Next, they conducted open coding to generate initial codes capturing recurring aims, practices and reasons related to the use of messaging platforms in PrEP care work. Through iterative discussions and constant comparison across transcripts, the research team examined relationships among codes and clustered them into candidate themes. These themes were then reviewed, refined, and reorganized through multiple rounds of team discussion to ensure internal coherence and clear distinctions between themes, as well as alignment with the research questions. Finally, the themes were named and defined to capture key patterns. This study was approved by an institutional research ethics review board. All participants provided informed consent, and all data were anonymized and de-identified during transcription and analysis.

3 RESULTS

3.1 What do PrEP navigators in Taiwan use LINE to do in their case management work?

PrEP navigators used LINE as an integral part of their everyday case management work outside the clinic. Two primary forms of work were carried out: reaching clients for routine follow-up and responding to emergent concerns in everyday life.

3.1.1 Reaching Clients for Appointments and Administrative Follow-up. Navigators used LINE to proactively reach clients for scheduled clinic visits and required administrative follow-up. By monitoring clients' visit histories through internal databases, navigators identified those approaching their next recommended appointment and sent reminders via LINE. As P5 explained: "We have our own database to manage people who are using PrEP. When a client comes in for a visit, we record the visit date and then calculate when their next visit should be. We look at who needs to come back next week or in two weeks, and then we use LINE to remind them—like, 'Hey, do you want to come back?' or 'Hey, it's time for you to come back.'"

In addition to appointment reminders, navigators relied on LINE to maintain contact with clients for administrative follow-up tasks, such as completing routine questionnaires or assessments. As P3 noted: "We use LINE to stay in contact with clients so that we can remind them to fill out follow-up forms or come in for assessments."

3.1.2 Responding to Emergent Concerns Outside the Clinic. Navigators also used LINE to respond to emergent concerns outside the clinic, particularly related to medication use and unexpected sexual situations. These interactions often required navigators to assess risk, ask follow-up questions, and provide situational guidance as circumstances unfolded. As P4 described: "Quite often, we respond to questions on LINE about taking PrEP incorrectly or not knowing how to take it. For example, a client once texted me, 'Navigator, I already took two PrEP pills, but on the day I went to meet someone, I realized I wasn't actually attracted to him at all. Do I still need to take the remaining two pills?' I asked him if there was sexual activity, and he said no. So I told him, 'If there was no sexual activity, you don't need to take PrEP.' But later, the same client came back and texted, 'Navigator, what should I do! We went out just planning to cuddle, but in the middle of it, he went ahead and we had sex.' [...] This client was originally using on-demand PrEP, but after learning about these situations, I discussed with him whether switching to daily PrEP might be a better option."

In other cases, navigators used LINE to guide clients through urgent decision-making after missed or delayed doses. As P1 explained: "Most of the time on LINE, I'm answering medication-related questions. Clients might miss a dose or take it a few hours late, and they'll ask what to do. If the sex was condomless or the condom broke, I'll tell them they need PEP (another type of medicine). I'll help them figure out when the nearest clinic is open. I'll think

through the steps with them one by one, so they feel more reassured and have a clear process to follow."

3.2 What affordances of LINE support these care management practices?

Across both forms of work, PrEP navigators repeatedly drew on a set of interface affordances provided by LINE. Rather than mapping onto a single task, these affordances were flexibly mobilized to support different aspects of case management.

3.2.1 Universality: Making Clients Reachable Over Time. The widespread adoption of LINE was a key reason navigators selected it as the primary platform for case management communication. Navigators emphasized that relying on a mainstream messaging application was essential for maintaining contact with clients over time. As P2 explained: "Everyone uses LINE. I always choose a platform that everyone is already using to communicate. I can't really use something like WhatsApp, because not that many people use it. When I choose a platform, what matters most to me is whether it's mainstream and something everyone already uses."

Similarly, navigators described intentionally offering LINE as a dedicated channel for PrEP-related communication in order to reduce barriers to contact. As P5 noted, "We provide a LINE channel specifically for PrEP clients so they can contact us one-on-one, because LINE is a messaging app that almost everyone uses."

3.2.2 Real-time Multimodality: Reducing Uncertainty and Supporting Care Across Situations. LINE's support for multiple communication modes allowed navigators to adapt communication to different situations and needs. Across administrative coordination, medical assessment, and emotional support, navigators shifted between visual, text-based, and voice communication to reduce uncertainty and respond more effectively.

Visual communication supported case management by providing concrete evidence and additional context. Navigators used images sent from the clients to verify the completion of administrative requirements, such as CDC (Taiwan Centers for Disease Control) questionnaires. As P4 described: "After clients join the CDC program, they're required to submit questionnaires and report whether they've completed anonymous testing. After they finish, I ask them to send me a screenshot showing the questionnaire is completed, so I can confirm that they've actually done it."

Images were also used to support medical assessment outside the clinic. When clients experienced possible medication side effects, such as allergic reactions, photos provided additional cues beyond text descriptions. As P3 noted: "Some clients experience drug allergies. They describe their allergic reactions to me on LINE, and some of them even take photos of skin rashes and send the pictures through LINE so we can understand what's going on."

Text-based communication allowed navigators to shape interactions in ways that conveyed attentiveness and relational warmth. Navigators emphasized they deliberately provided personalized replies to clients. As P2 explained: "Clients don't like template-style replies. What they want is an actual conversation with me. I think the most important thing between a navigator and a client is building a relationship. If a client asks a question and I send a canned response, they won't want to read it. They feel like they're

talking to something cold and lifeless. So I always talk to them conversationally, explaining things step by step, like chatting."

Navigators also used linguistic cues and forms of address to make text-based interactions feel more human. P4 described incorporating sentence-final particles common in Taiwanese communication, while P5 noted using terms such as "Dear" and adding small situational remarks to personalize routine exchanges. P4 reflected: "I think one very distinctive part of Taiwanese communication culture is the use of utterance-final particles like 'a', 'la', or 'ne'. When I include these particles in text, at least it makes clients feel the interaction is more friendly, more human. In my first two or three years on the job, my replies were fast, blunt, and precise because I had so many LINE messages to answer. But in the past two years, I've realized that clients really do feel the difference." P5 described: "When a client asks on LINE whether they can reschedule an appointment, I'll reply 'Oh sure,' and I often add 'Dear,' like 'Dear [name]'. When the weather was colder, I would also remind them in the message, like, 'Hey, make sure you dress warmly when you come in for your visit,' so they don't feel like the interaction is cold or impersonal."

Lastly, **Voice communication** was used when text-based interaction was insufficient to address clients' emotional states or urgent concerns. Navigators described shifting from typing to voice calls, enabled by LINE's built-in free calling feature, when clients were overwhelmed by anxiety and unable to process written information. As P4 explained: "A client was feeling very anxious and asked me a series of questions on LINE. I initially replied by typing, but the client couldn't really understand because they were under a lot of emotional stress at the time. So I asked the client whether we could talk on the phone instead."

Beyond conveying information, voice calls also played a role in providing reassurance. As noted in P2, clients sometimes called simply to hear the voice of a navigator during moments of distress: "Clients need an outlet to release their anxiety and unease in that moment. Sometimes they call me just to hear my voice. I think what clients care about is not only getting information, but they also want to build a relationship with me through this, so they can feel more reassured."

3.2.3 Asynchrony: Balancing Responsiveness, Non-intrusion, and Workload. LINE's asynchronous communication supported both follow-up and responsiveness while allowing navigators to manage timing. When reminding clients about appointments or administrative tasks, navigators could send messages without interrupting clients' daily lives, enabling clients to respond to the reminders at their own pace. As P6 explained, "LINE is basically like free text messaging. Clients can read our messages and even choose not to reply. The point is that we want to remind them in a way that doesn't interrupt or interfere with their daily lives."

At the same time, navigators relied on asynchrony to manage their own availability and workload. As P2 described, urgent issues were handled immediately, while less time-sensitive questions were deferred: "I'm basically a 24-hour navigator. If it's something urgent that needs immediate attention, like forgetting to take PrEP, I'll reply right away. But if it's something that can wait, I'll reply the next day. For example, questions about the program itself can usually wait until the next day."

In addition, navigators used asynchronous messaging to structure interactions more. As P1 noted, this allowed navigators to review messages when time permitted and respond more directly: “I always tell my clients that if they have any questions about taking PrEP, they can message me on LINE but they need to write everything clearly. That way, when I finish my work and see all the information, I can respond directly.”

3.2.4 Private One-on-One Space: Enabling Disclosure and Relational Feedback. The one-on-one chat format on LINE created a private space that extended interaction beyond the clinic. Navigators contrasted this channel with in-clinic encounters and group-based messaging environments, noting that privacy shaped what clients were willing to ask and share. As P3 explained: “In a typical medical setting, patients can only interact with doctors or ask questions during the clinic visit itself. But our LINE communication is all one-on-one chats, so clients can ask questions anytime, and they’re more willing to ask private or sensitive questions. If it were a group chat, clients would know others could see their messages, and they would limit what they choose to say.”

This private space also allowed questions to emerge after clients left the clinic, particularly when anxiety made it difficult to engage during in-person encounters. As P4 noted: “In the clinic, some clients are too anxious to really take in my health education. So I tell them, ‘If you’re still very worried or anxious, and you feel like your questions weren’t fully answered here, you can message me on LINE when you get home if more questions come up.’ And afterward, he really did message me on LINE with many questions.”

In addition to facilitating disclosure, LINE acted as a channel enabled clients to provide relational feedback after clinical encounters. As P1 mentioned: “Many first-time clients, after leaving our clinic, will immediately message me on LINE and say, ‘I felt really comfortable during today’s visit. I think this environment is very LGBTQ-friendly.’”

4 DISCUSSION

Our findings show that PrEP navigators invest substantial time and effort to support client engagement and medication adherence. LINE playing a critical role here as it enables navigators to maintain contact, respond to emergent concerns, and sustain relationships over time, allowing preventive care to be integrated into daily routines rather than confined to episodic clinic visits. In this sense, technology enables preventive care to function as an ongoing, relational practice rather than merely increasing efficiency.

However, this technological support entails less visible costs. The same affordances that extend care beyond the clinic also blur temporal and relational boundaries for navigators, as illustrated by P2, who described being effectively available around the clock to respond to medical questions and related emotional distress. While asynchronous messaging reduces intrusion for clients, it extends the presence of care work into navigators’ everyday lives, rendering relationship-building both a resource and a form of labor. Similar tensions have been observed in other care settings, where instant messaging extends care coordination while simultaneously increasing expectations of constant availability and time work for care workers [8]. This phenomenon aligns with recent CSCW discourse on how digital transformation can inadvertently exploit care

as a non-material resource [9]. Together, these dynamics produce paradox that is difficult to disentangle: technological convenience can increase the burden of care on navigators [10], even as their willingness to absorb this burden may enable clients to feel safe, accepted, and motivated to continue preventive self-care.

From this perspective, future digital support tools, such as AI-powered chatbots, should be approached cautiously. The uncritical deployment of these technologies may lead to trust decay and cannot replace the practical wisdom required for clinical judgment [22]. A key challenge lies in determining which aspects of care can be delegated to automated systems without undermining trust, reassurance, and engagement. Prior work shows that while caregivers value technology handling mundane tasks, they prioritize preserving meaningful emotional interactions for human connection [18]. Substituting human presence with “artificial care” risks diminishing the authenticity of connections, particularly when users perceive responses as scripted, misattuned, or pitying [18, 32]. Accordingly, researchers have argued for a human-in-the-loop approach, in which backend workflows are automated while emotional engagement is reserved for human workers in order to maintain the “human touch” [10, 30].

Building on these insights, we propose directions for future design research grounded in local practices and expectations: **(1) Designing for relational communication:** How might conversational designs better support the micro-level textual choices (wording, tone, and culturally specific cues) through which navigators convey warmth and human presence? Prior work suggests that aligning AI communication styles with users’ cultural backgrounds can shape how such systems are perceived [3, 31]. **(2) Supporting asynchronous rhythms without eroding boundaries:** How might systems make availability and response timing legible (for example, expected reply windows) without feeling rejecting or overly formal? What lightweight triage cues can help distinguish routine follow-up from urgent concerns while preserving conversational flow? More broadly, how could technologies help workers “tame” workflows rather than be tamed by boundless accessibility [8]? **(3) Facilitating modality switching and handoff:** What signals should trigger escalation from chat to calls or in-person care, and how can systems make that transition easy in-the-moment? How can consent-aware handoffs preserve continuity (for example, minimal summaries) so clients do not need to repeat sensitive histories, while still benefiting from multimodal communication for accurate understanding and care coordination [15]? **(4) Preserving privacy and a sense of safety:** What safety-by-default visibility controls (for example, hidden previews, neutral notifications) can reduce inadvertent disclosure in stigmatized-care messaging? How can systems support quick “is it safe to talk now?” checks and lower-visibility modes when clients are in shared-device or public settings, while still enabling sensitive disclosure and relational feedback?

5 CONCLUSION

This case study examined how PrEP navigators in Taiwan use LINE to support case management and how the platform’s affordances shape care work. Our findings show that LINE is deeply embedded in everyday practices of follow-up, response, and relationship maintenance, enabling preventive care to be enacted as an ongoing

process beyond clinic visits. We further highlight how universality, multimodality, asynchrony, and private one-on-one interaction are flexibly mobilized across care situations, and propose directions for future design based on these findings.

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References

- [1] Abhishek Aggarwal, Cheuk Chi Tam, Dezhi Wu, Xiaoming Li, and Shan Qiao. 2023. Artificial intelligence-based chatbots for promoting health behavioral changes: systematic review. *Journal of medical Internet research* 25 (2023), e40789.
- [2] Jared M Baeten, Deborah Donnell, Patrick Ndase, Nelly R Mugo, James D Campbell, Jonathan Wangisi, Jordan W Tapper, Elizabeth A Bukusi, Craig R Cohen, Elly Katabira, et al. 2012. Antiretroviral prophylaxis for HIV prevention in heterosexual men and women. *New England journal of medicine* 367, 5 (2012), 399–410.
- [3] Rebekah Lee Baik, Stephanie Lee, Serena Jinchun Xie, Wang Liao, Elina H Hwang, and Weichao Yuwen. 2025. Adapting communication styles in health chatbot using large language models to support family caregivers from multicultural backgrounds. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [4] Katie B Biello, Salvatore R Daddario, Jonathan Hill-Rorie, Donna Futterman, Patrick S Sullivan, Lisa Hightow-Weidman, Jeb Jones, Matthew J Mimiaga, and Kenneth H Mayer. 2022. Uptake and acceptability of MyChoices: results of a pilot RCT of a mobile app designed to increase HIV testing and PrEP uptake among young American MSM. *AIDS and behavior* 26, 12 (2022), 3981–3990.
- [5] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health* 11, 4 (2019), 589–597.
- [6] Anthony Duffy, Gregory J Christie, and Sylvain Moreno. 2022. The challenges toward real-world implementation of digital health design approaches: narrative review. *JMIR Human Factors* 9, 3 (2022), e35693.
- [7] Elissa Z Faro, Joanne E Mantell, Tatiana Gonzalez-Argoti, Susie Hoffman, Zoe Edelstein, Benjamin Tsoi, and Laurie J Bauman. 2022. Implementing PrEP services in diverse health care settings. *JAIDS Journal of Acquired Immune Deficiency Syndromes* 90, S1 (2022), S114–S128.
- [8] Helena Hirvonen, Mia Tammelinn, Antti Hämmäläinen, and Sakari Taipale. 2022. Group-based instant messaging in Finnish residential elder care work: Taming the technology or vice versa? *New Technology, Work and Employment* 37, 2 (2022), 231–249.
- [9] Azra Ismail, Deepika Yadav, Meghna Gupta, Kirti Dabas, Pushpendra Singh, and Neha Kumar. 2022. Imagining caring futures for frontline health work. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–30.
- [10] Naveena Karusala, Shirley Yan, Nupoor Rajkumar, Victoria G, and Richard Anderson. 2023. Speculating with Care: Worker-centered Perspectives on Scale in a Chat-based Health Information Service. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (2023), 1–26.
- [11] Theodoros Kesisidis and Raphael J Landovitz. 2011. Preexposure prophylaxis for HIV prevention. *Current HIV/AIDS Reports* 8, 2 (2011), 94–103.
- [12] Moh Heri Kurniawan, Hanny Handiyani, Tuti Nuraini, Rr Tutik Sri Hariyati, and Sutrisno Sutrisno. 2024. A systematic review of artificial intelligence-powered (AI-powered) chatbot intervention for managing chronic illness. *Annals of Medicine* 56, 1 (2024), 2302980.
- [13] Morgan LaBelle, Carol Strong, and Yuan-Chi Tseng. 2020. mHealth strategies to promote uptake and adherence to PrEP: a systematic review. In *International Conference on Human-Computer Interaction*. Springer, 99–113.
- [14] Moustafa Laymouna, Yuanchao Ma, David Lessard, Tibor Schuster, Kim Engler, and Bertrand Lebouché. 2024. Roles, benefits, and limitations of chatbots in health care: rapid review. *Journal of medical Internet research* 26 (2024), e56930.
- [15] Brenna Li, Amy Wang, Patricia Strachan, Julie Anne Séguin, Sami Lachgar, Karyn C Schroeder, Mathias S Fleck, Renee Wong, Alan Karthikesalingam, Vivek Natarajan, et al. 2024. Conversational AI in health: Design considerations from a Wizard-of-Oz dermatology case study with users, clinicians and a medical LLM. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–10.
- [16] Jay Chiehien Liao, Huei-Jiuan Wu, Tsan-Tse Chuang, Tsai-Wei Chen, and Carol Strong. 2025. Enhancing PrEP adherence through person-centred mobile app interventions: a real-world data and machine learning approach using UPrEP among gay, bisexual and other men who have sex with men in Taiwan. *Journal of the International AIDS Society* 28 (2025), e70033.
- [17] Albert Y Liu, Nicole D Laborde, Kenneth Coleman, Eric Vittinghoff, Rafael Gonzalez, Gretchen Wilde, Annie L Thorne, Ed Ikeguchi, Laura Shafner, Lauren Sunshine, et al. 2021. DOT diary: developing a novel mobile app using artificial intelligence and an electronic sexual diary to measure and support PrEP adherence among young men who have sex with men. *AIDS and Behavior* 25, 4 (2021), 1001–1012.
- [18] Jiaying" Lizzy" Liu, Shuer Zhuo, Xingyu Li, Andrew Dillon, Noura Howell, Angela DR Smith, and Yan Zhang. 2025. From Regulation to Support: Centering Humans in Technology-Mediated Emotion Intervention in Care Contexts. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (2025), 1–32.
- [19] LY Corporation. 2024. Continue to Evolve Without Fearing Failure: LINE Taiwan's Key to Success and Organizational Culture. <https://www.lycorp.co.jp/en/story/20250226/linetaiwan.html>. Accessed: 2025-01-21.
- [20] Julia L Marcus, Timothy Buisker, Tara Horvath, K Rivet Amico, Jonathan D Fuchs, Susan P Buchbinder, Robert M Grant, and Albert Y Liu. 2014. Potential interventions to support adherence to HIV preexposure prophylaxis (PrEP): a systematic review. *HIV medicine* 15, 7 (2014), 385.
- [21] DC Matthews, K McNeil, M Brillant, C Tax, P Maillet, CA McCulloch, and M Glogauer. 2016. Factors influencing adoption of new technologies into dental practice: a qualitative study. *JDR Clinical & Translational Research* 1, 1 (2016), 77–85.
- [22] Jaana Parviainen and Juho Rantala. 2022. Chatbot breakthrough in the 2020s? An ethical reflection on the trend of automated consultations in health care. *Medicine, health care and philosophy* 25, 1 (2022), 61–71.
- [23] Aly Passanante, Ed Pertwee, Leesa Lin, Kristi Yoonsup Lee, Joseph T Wu, and Heidi J Larson. 2023. Conversational AI and vaccine communication: systematic review of the evidence. *Journal of medical Internet research* 25 (2023), e42758.
- [24] Erika S Poole. 2013. HCI and mobile health interventions: how human-computer interaction can contribute to successful mobile health interventions. *Translational behavioral medicine* 3, 4 (2013), 402–405.
- [25] Parya Saberi, Kristin Ming, Hyman Scott, Albert Liu, and Wayne Steward. 2020. "You can't have a PrEP program without a PrEP Coordinator": Implementation of a PrEP panel management intervention. *PLoS One* 15, 10 (2020), e0240745.
- [26] Carol Strong, Huei-Jiuan Wu, Yuan-Chi Tseng, Chien-Wen Yuan, Yi-Fang Yu, Jay Chiehien Liao, Yi-Wen Chen, Yi-Chen Hung, Chia-Wen Li, Po-Hsien Huang, et al. 2020. Mobile app (UPrEP) to monitor adherence to pre-exposure prophylaxis in men who have sex with men: protocol for a user-centered approach to mobile app design and development. *JMIR Research Protocols* 9, 12 (2020), e20360.
- [27] Taiwan Centers for Disease Control. 2023. PrEP Program Services. Taiwan CDC website. <https://www.cdc.gov.tw/En/Category/QAPage/pzjkJa4dgVjdk6emQjM-3g> Last updated 2023-01-16. Accessed 2026-02-23.
- [28] Gareth Terry, Nikki Hayfield, Victoria Clarke, Virginia Braun, et al. 2017. Thematic analysis. *The SAGE handbook of qualitative research in psychology* 2, 17-37 (2017), 25.
- [29] Michael C Thigpen, Poloko M Kebaabetswe, Lynn A Paxton, Dawn K Smith, Charles E Rose, Tebogo M Segolodi, Faith L Henderson, Sonal R Pathak, Fatma A Soud, Kata L Chillag, et al. 2012. Antiretroviral preexposure prophylaxis for heterosexual HIV transmission in Botswana. *New England journal of medicine* 367, 5 (2012), 423–434.
- [30] Yuan-Chi Tseng, Samuel Chen, Kang-Heng Mah, and Yu-Chang Chen. 2025. Designing an AI Chatbot for Team-Based Diabetes Care: An Iterative Human-in-the-Loop Approach. In *International Conference on Human-Computer Interaction*. Springer, 261–276.
- [31] Yuan-Chi Tseng, Weerachaya Jarupreechachan, and Tuan-He Lee. 2023. Understanding the benefits and design of chatbots to meet the healthcare needs of migrant workers. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (2023), 1–34.
- [32] Cai-Ling Wang and Yuan-Chi Tseng. 2024. "I Don't Want to be Pitied by a Bot": Understand How to Design Chatbots to Support People Being Ghosted on Dating Applications. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [33] Huei-Jiuan Wu, Yi-Fang Yu, Stephanie Wen-Wei Ku, Yuan-Chi Tseng, Chien-Wen Yuan, Chia-Wen Li, Po-Hsien Huang, Nai-Ying Ko, Peter L Anderson, and Carol Strong. 2022. Usability and effectiveness of adherence monitoring of a mobile app designed to monitor and improve adherence to event-driven and daily HIV pre-exposure prophylaxis among men who have sex with men in Taiwan. *Digital health* 8 (2022), 20552076221102770.
- [34] Farzaneh Yousefi, Florian Naye, Steven Ouellet, Achille-Roghemrazangba Yameogo, Maxime Sasseville, Frédéric Bergeron, Marianne Ozkan, Martin Cousineau, Samira Amil, Caroline Rhéaume, et al. 2025. Artificial intelligence in health promotion and disease reduction: rapid review. *Journal of medical Internet research* 27 (2025), e70381.
- [35] Chien Wen Yuan, Yuan-Chi Tseng, and Carol Strong. 2023. Understanding and designing multi-level preventive medication support against HIV for men who have sex with men in Taiwan. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (2023), 1–30.